

Indian Institute of Technology (Indian School of Mines) Dhanbad

Monsoon Mid Semester Question Paper 2025-26

Examination: B.Tech (1st Year Common)
Discipline: Mechanical Engineering
Subject: Manufacturing Process (NMES 101)

Time: 1 Hour
Total Marks: 30
Session: 2025 - 26

Part - I (Casting - 10 marks)

1. A 120 mm long and 30 mm diameter cylindrical component is desired. The casting process used is sand casting, and after casting, the component is machined to improve surface finish by removing a 1.5 mm thick layer from every side. The solidification shrinkage of the metal is 6%. Calculate the desired dimensions of the wooden pattern used in sand casting considering both machining and shrinkage allowances. [3]
2. A spherical-shaped casting solidifies in 24 minutes. Another casting of the same material is in the shape of a cylinder with the same height and diameter as the diameter of the sphere. Estimate the solidification time for the cylindrical casting. [7]

Part - II (Forming - 10 marks)

3. Write four differences between hot metal working and cold metal working in tabular form. [4]
4. In a tensile test, two pairs of stress-strain values were measured for the specimen after yielding: (1) true stress = 217 MPa, and true strain = 0.35; and (2) true stress = 259 MPa, and true strain 0.68. Based on these data points, determine the strength coefficient and strain-hardening exponent. [6]

Part - III (Metrology - 10 marks)

5. Write four Indian Standards (IS) and their corresponding names/titles. [2]
6. A slip gauge of 20 mm length (True value) was used to determine precision and accuracy of mechanical vernier micrometer having range 0 - 25 mm and least count 0.001 mm after setting zero error. Readings are 20.001 mm, 19.997 mm, 20.003 mm, 19.998 mm, and 20.000 mm respectively. Find σ_T and $\bar{\sigma}_X$. [3]
7. A ground steel surface is inspected using a stylus profilometer. The instrument has a cut-off length of 0.8 mm, and the total evaluation length is 4.0 mm. The recorded deviations (in μm) from the mean line at 0.2 mm intervals are:
+1.2, -0.8, +1.5, -1.0, +2.1, -1.8, +1.6, -1.4, +1.9, -0.9, +1.7, -1.3, +2.0, -1.6, +1.4, -1.1, +1.8, -1.7, +1.5, -1.2
a) Draw the waviness profile with the mean line.
b) Compute the overall R_a for the surface.
c) Compute the Root Mean Square Roughness (R_q) for the entire 4.0 mm profile.
d) Determine the ten-point height roughness (R_z), defined as the average of the five highest peaks and five deepest valleys.
e) The drawing specification requires $R_a \leq 1.6 \mu\text{m}$ and $R_z \leq 2 \mu\text{m}$.
o Does the surface pass inspection? Justify. [5]

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